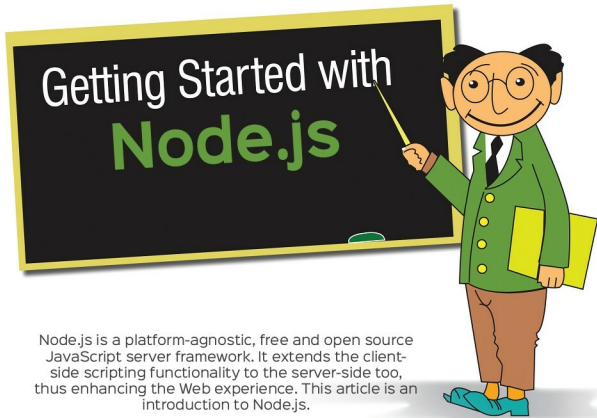




Node.js is a platform-agnostic, free and open source JavaScript server framework. It extends the client-side scripting functionality to the server-side too, thus enhancing the Web experience. This article is an introduction to Node.js.

If you've ever used a search box on a website, checked the live score for a soccer game, or streamed a video, you probably have JavaScript to thank for making your experience seamless.

JavaScript (JS) is a high-level, interpreted programming language. It is often identified as dynamic, weakly-typed, prototype-based and multi-paradigm. Simply put, it is the language that infuses life into Web pages, providing animations, visuals and a degree of interactivity to the rather static elements of Web design. It was designed as a text-based language that could run as part of a Web-based application. While the name, syntax and structure seem similar to a lot of constructs available in Java, the design varies greatly with JavaScript influenced by languages such as Self and Scheme. Originally intended to complement Java, it grew exponentially to form one of the three pillars of Web development—the others being HTML and CSS. Today, almost all modern browsers contain a JS engine to support the use of JavaScript without the need for extensions, which is a testimony to its popularity.

JavaScript faced a fair amount of controversy over the course of multiple releases between 1996 and 2009 before all parties finally agreed to implement it as a common

standard—thus, ECMAScript 5 was born. Meanwhile, open source efforts consolidated around AJAX and other libraries built to complement Vanilla JS (as 'plain JavaScript code' is commonly referred to), leading to the birth of the community of developers that continues to contribute to it today.

Why use Node.js?

JavaScript was primarily meant to be implemented by client-side scripting—embed the script in HTML and let the JS engine within the user's Web browser handle the rest. However, Node.js extends the functionality to server-side scripting, representing the 'JavaScript everywhere' paradigm. By using Node.js, developers can generate dynamic Web pages before sending them to the user. In this manner, it unifies Web application development by using a single programming language for the client and server.

The motivation to develop this framework for JavaScript was to allow for a system designed for real-time communication from the ground up, rather than one that offers support for it as an additional feature. Node.js is built on an event-driven architecture capable of performing asynchronous I/O. Such design choices aim to optimise